

Aanya A. Gupta

aanya.gupta30@duke.edu | [linkedin.com/in/aanyagupta1](https://www.linkedin.com/in/aanyagupta1)

EDUCATION

Duke University

Electrical and Computer Engineering

Durham, North Carolina

Expected Graduation May 2030

- **Relevant Coursework:** Data Structures/Computer Algorithms, Differential Equations, Chem/Biology, Engineering Design

Thomas Jefferson High School for Science and Technology (TJHSST)

Advanced Studies Diploma

Alexandria, Virginia

June 2026

- **Cumulative GPA:** 4.60 (W)
- **Relevant Coursework:** Statistics, Multivariable Calculus, Linear Algebra, AP Physics, Computer Vision, Mobile/Web Dev
- **Relevant Extracurricular Activities:** TJ Climbing NGO (Founder), TJHSST TEKNOS Research Journal Club (Associate Editor-in-Chief), TJHSST Coding Lady Colonials Club (President), HackTJ Hackathon (President/Director)

WORK EXPERIENCE

Raytheon Summer Internship

Software and Research Intern (Summer 2027 Return Offer Received)

Arlington, Virginia

June 2026 — August 2026

- Rebuilt a fragmented benchmarking pipeline into a modular, end-to-end evaluation framework in Python for a fraud-detection research program, reducing per-run setup from 4 hours to 10 minutes and making results reproducible across 5 data sources
- Benchmarked the robustness of existing anomaly-detection models against realistic synthetic event data, characterizing 3 failure modes where detectors missed suspicious activity; the test harness was adopted by a 35-person team and 3 customers

Agent-Based Modeling Research under Dr. Taylor Anderson

Research Intern

Fairfax, Virginia

May 2024 — Present

- Engineered an agent-based model of vaccine uptake across 5.5K synthetic agents linking social factors and health policy
- Pioneered a novel approach to small-area estimation using iterative proportional fitting, improving the accuracy of local health outcome predictions by 2% over established CDC benchmarks across 50+ counties
- Co-authored multiple publications and open-sourced model code and datasets for public health research and decision-making

Givology (Global Education 501(c)(3))

Head of Events and Special Projects

Fairfax, Virginia

September 2022 — Present

- Spearheaded 6 fundraising campaigns, raising over \$18K to support education initiatives for children in developing countries
- Managed Facebook and LinkedIn profiles with 7.8K followers; created posts to highlight partner organizations

PROJECTS

MIT Beaver Works Summer Institute (BWSI)

Exact Ising Model Simulation

Cambridge, Massachusetts

June 2025 — August 2025

- Built an exact quantum simulation of the 1D transverse-field Ising spin chain in Q# and Qiskit; published on GitHub
- Applied quantum algorithms (Quantum Fourier Transform, Shor's) to optimize circuit efficiency; presented to 500+ peers

Stanford Clinical Science, Technology and Medicine Internship

AIDKnIT

Stanford, California

June 2024 — July 2024

- Developed AIDKnIT, an AI-driven robotics concept for patient-centered care, with the Stanford medX team and clinicians
- Produced and presented an ePatient case study to clinicians and engineering mentors

HackTJ Hackathon

SwiftStudy

Tysons, Virginia

March 2023

- Built SwiftStudy, an app converting uploaded notes into source-linked study questions (OpenAI API, AWS); 1st of 520

PEER-REVIEWED PUBLICATIONS

- Von Hoene, E., **Gupta, A.**, Kavak, H. et al. Evaluation of a spatial microsimulation framework for small-area estimation of population health outcomes using the behavioral risk factor surveillance system. *Int J Health Geogr* (2026). <https://doi.org/10.1186/s12942-026-00485-3>
- Von Hoene, E., **Gupta, A.**, et al. A Synthetic Population and Small-Area Estimate Dataset for Health Behaviors and Outcomes in the United States. *Nature Scientific Data*.
- **Gupta, A.**, Péter, S. A., et al. Geographically Weighted Surrogate Models for Rapid Small-Area Chronic Disease Estimation. *Applied Geography*.
- Von Hoene, E., **Gupta, A.**, et al. Evaluating the Transferability of a Synthetic Population Generation Approach for Public Health Applications. *Winter Simulation Conference 2025*. <https://doi.org/10.1109/WSC68292.2025.11338931>

TECHNICAL SKILLS

- Programming & Software: Python, C, C++, Java, R, Q#, Qiskit, HTML/CSS, Git/GitHub
- Data & Analytics: Statistical modeling, data visualization (Tableau, Power BI), Google Analytics
- Engineering Tools: AutoCAD, Fusion 360, Onshape

HONORS & INTERESTS

Honors: Congressional Award Gold Medalist, International Research Olympiad Semifinalist, 1st Place Continental Math League

Interests: Rock Climbing, Baking, Traveling, Anything Dogs